

From Hallucination to Justification

Hands-On Explainability for LLMs

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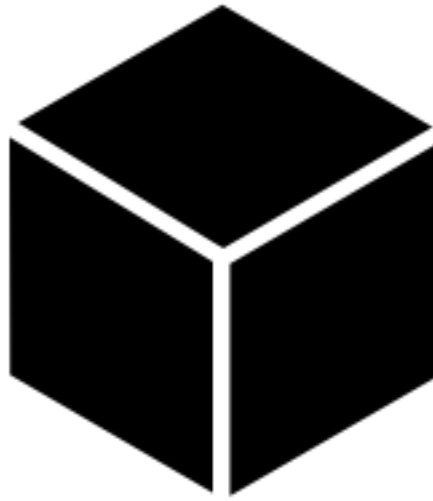
XAI



Explainable Artificial Intelligence (XAI) is the ability of AI systems to provide **clear and understandable explanations for their actions and decisions**, [...] to make the behavior of these systems understandable to humans by elucidating the **underlying mechanisms of their decision-making processes**.

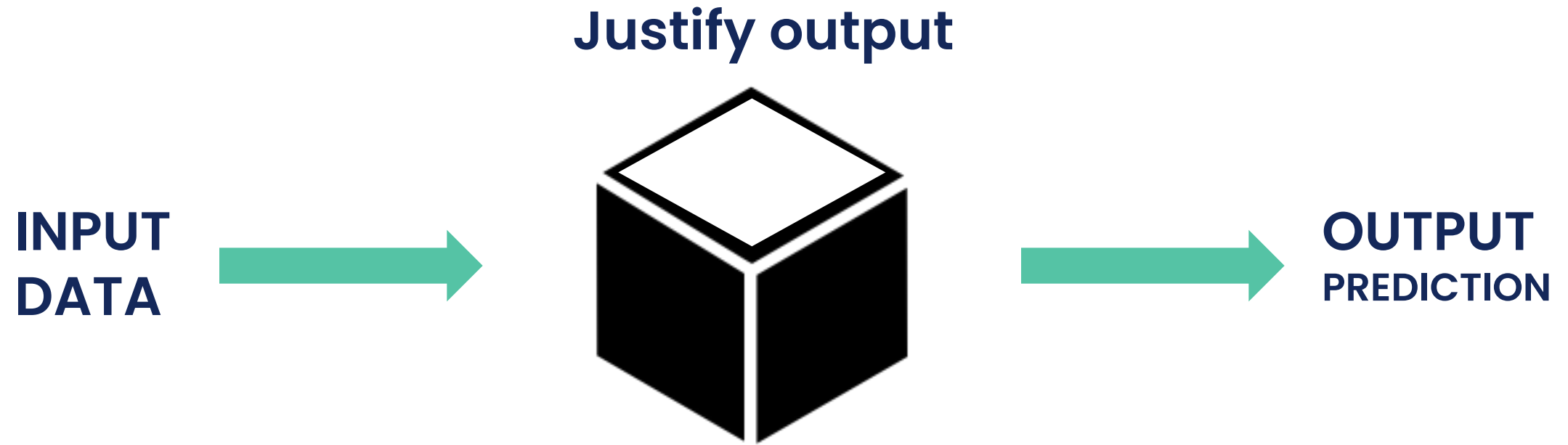
EDPS – European Data Protection Supervisor

**INPUT
DATA**

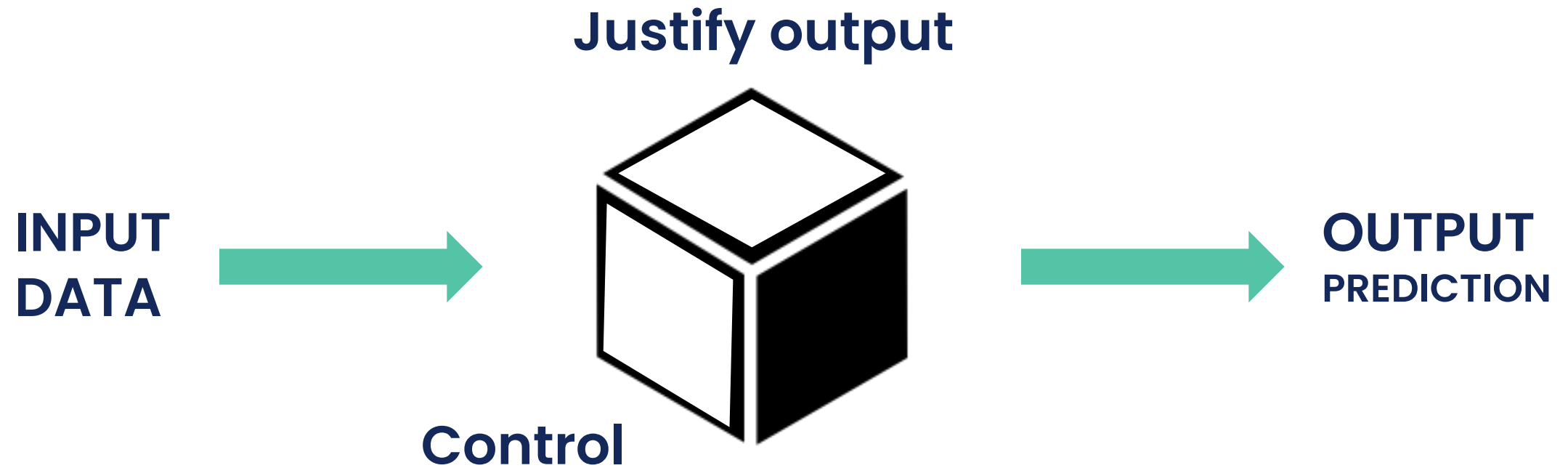


**OUTPUT
PREDICTION**

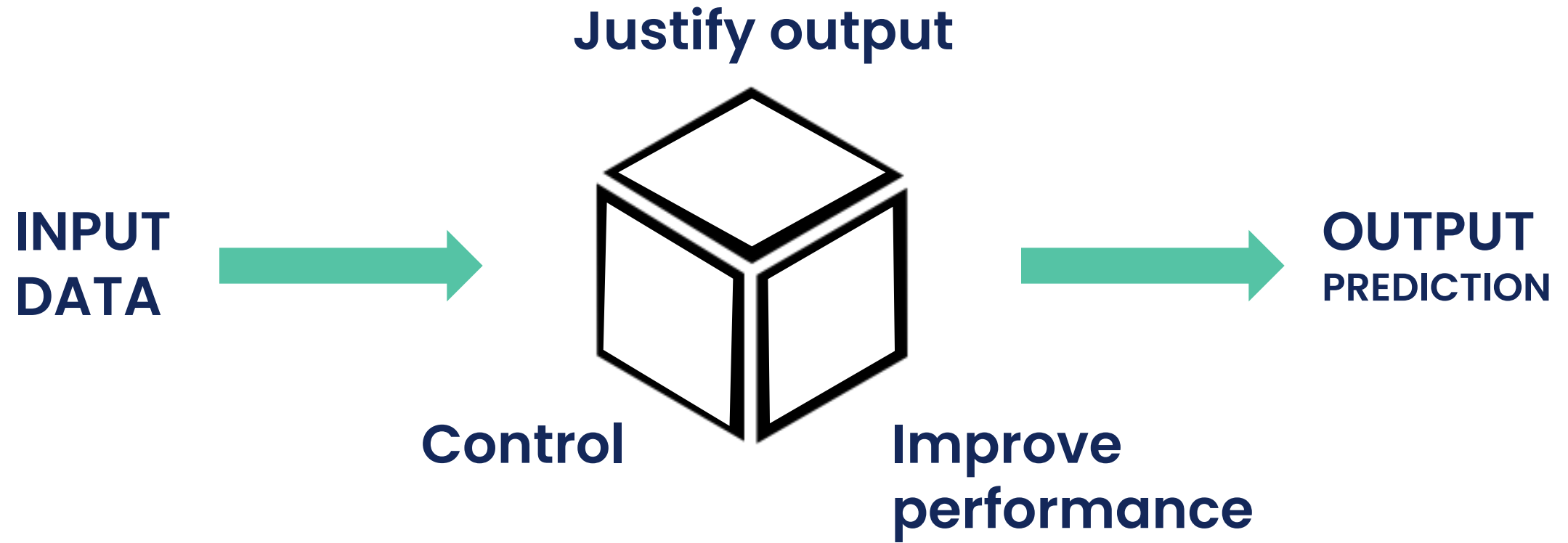
Benefits of XAI



Benefits of XAI



Benefits of XAI



Apple Card is accused of gender bias. Here's how that can happen

By Evelina Nedlund, CNN Business

🕒 5 min read · Updated 2:04 PM EST, Tue November 12, 2019

The New York Department of Financial Services is looking into the allegations of gender discrimination against users of the Apple Card. The allegations blew up on Twitter Saturday after tech entrepreneur David Heinmeier Hansson wrote that Apple Card offered him 20 times the credit limit as his wife, although they have shared assets and she has a higher credit score.

December 9th, 2025 | 6 min read

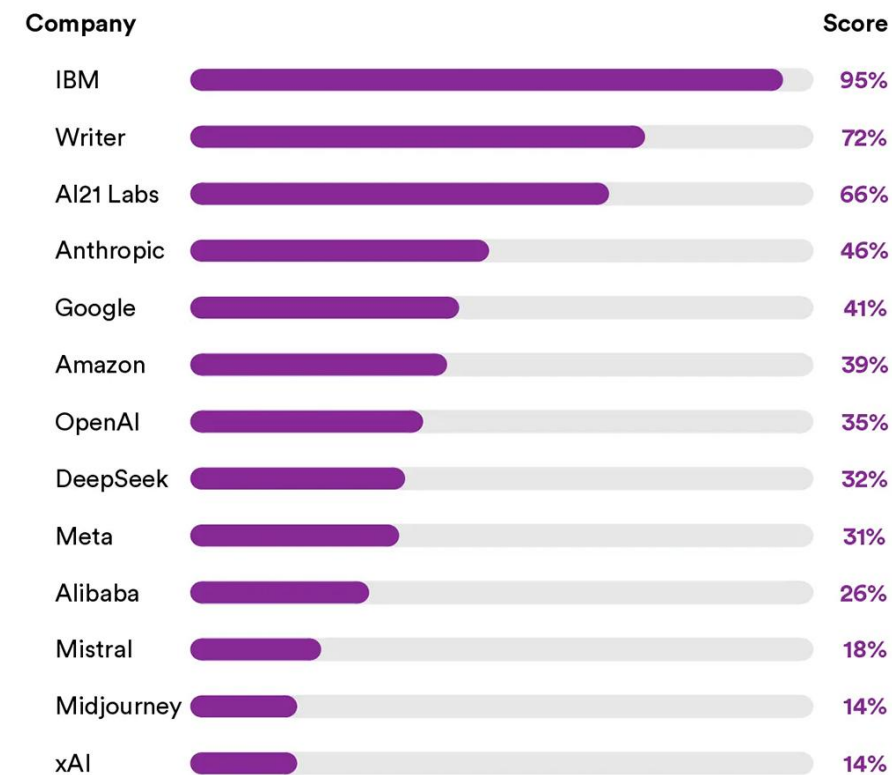
Artificial Intelligence

Transparency in AI is on the decline

A new analysis finds that AI companies now average just 40 out of 100 on transparency, marking a significant decline from last year.

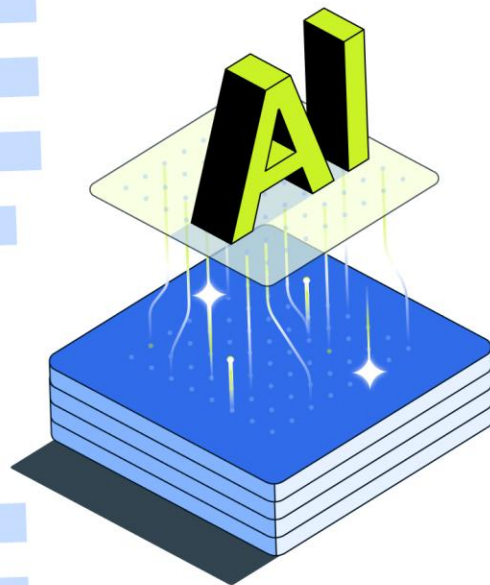
Foundation Model Transparency Index Total Scores, 2025

Source: 2025 Foundation Model Transparency Index



BREAKING NEWS

EU AI Act

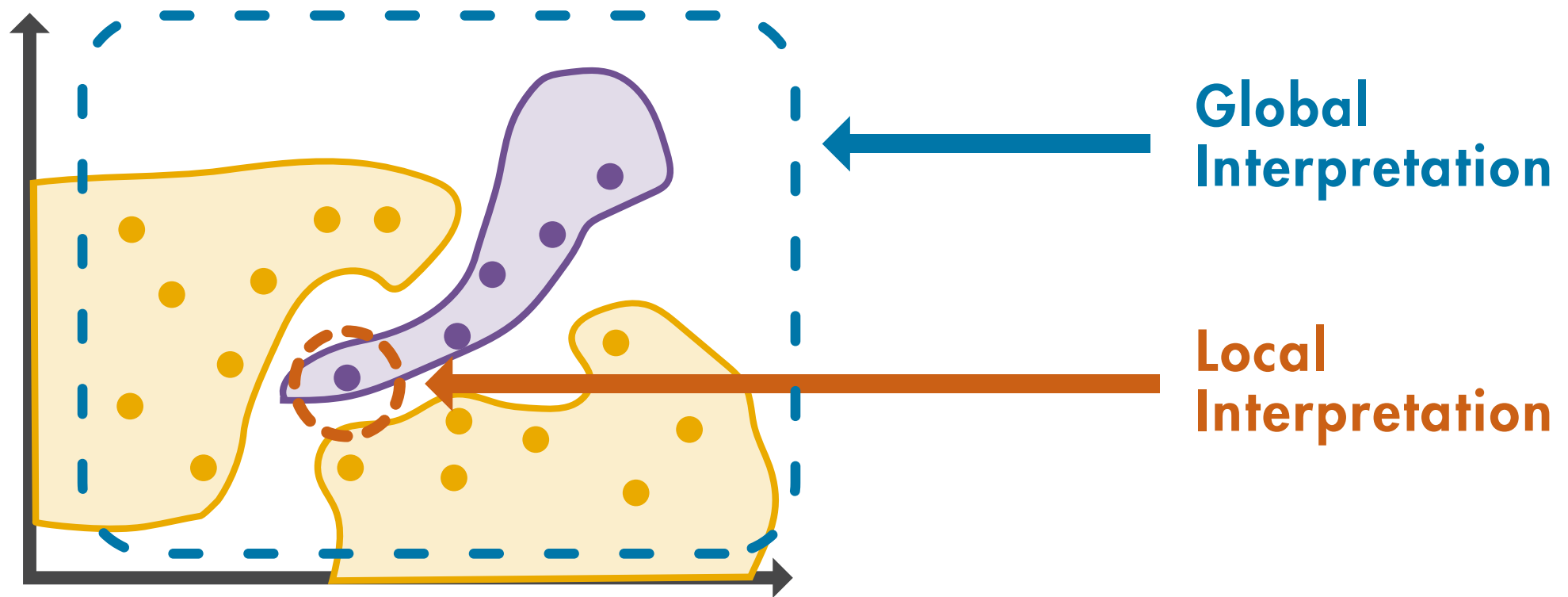


Article 13 (1) of the EU AI Act specifies:

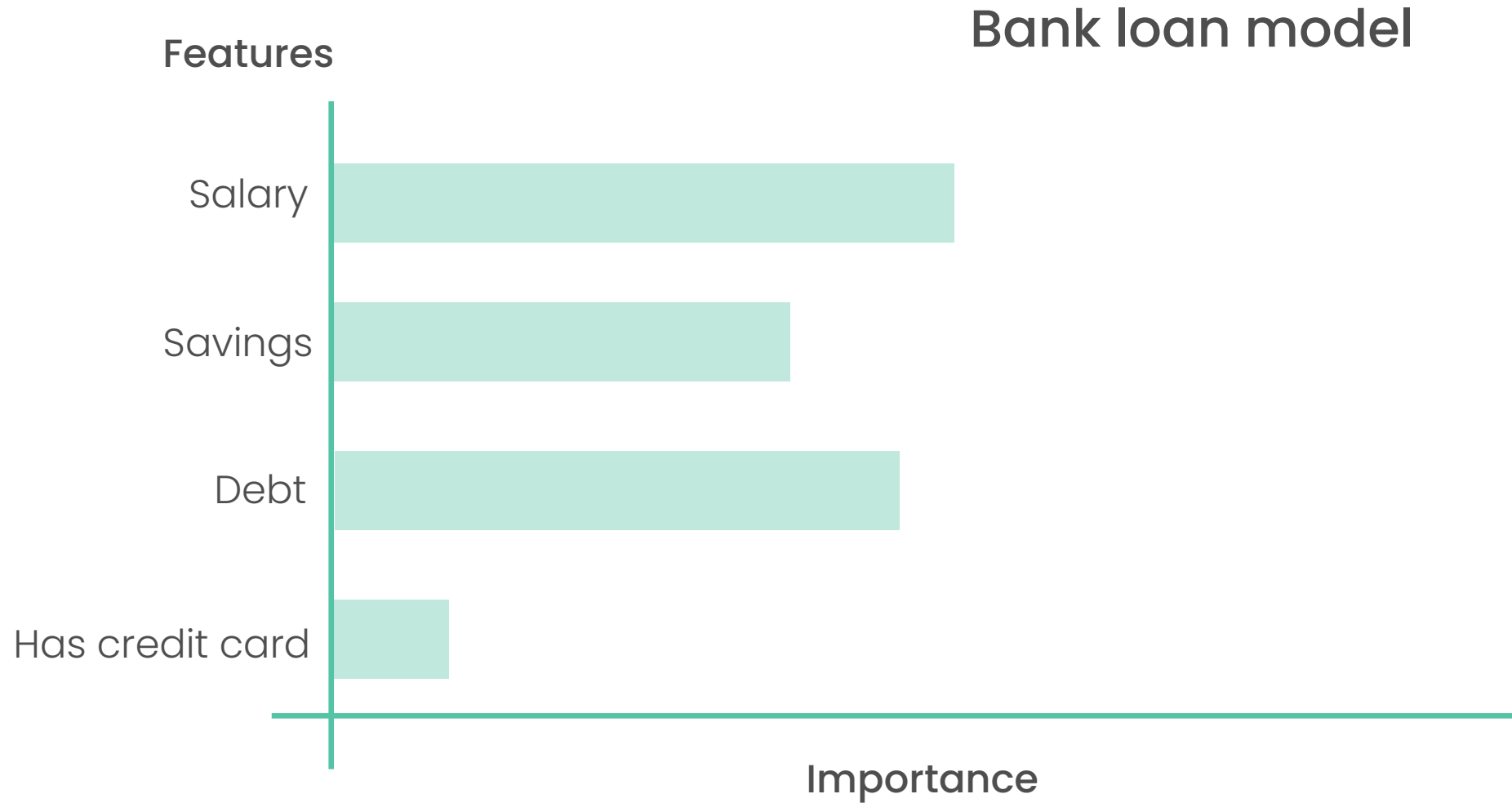
*'High-risk **AI systems shall be designed** and developed in such a way to ensure that their operation is sufficiently transparent **to enable users to interpret the system's output** and use it appropriately.'*



XAI 101

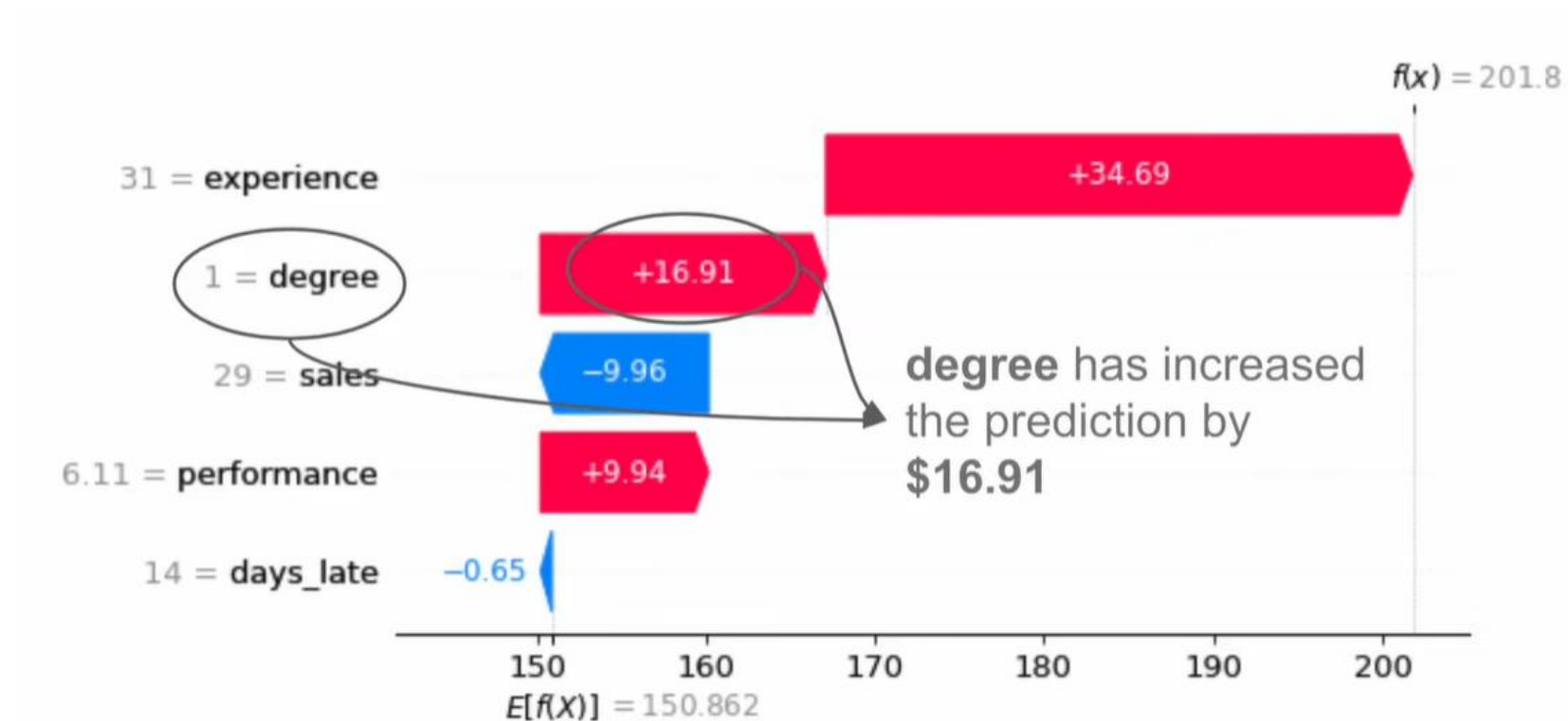


Feature importance



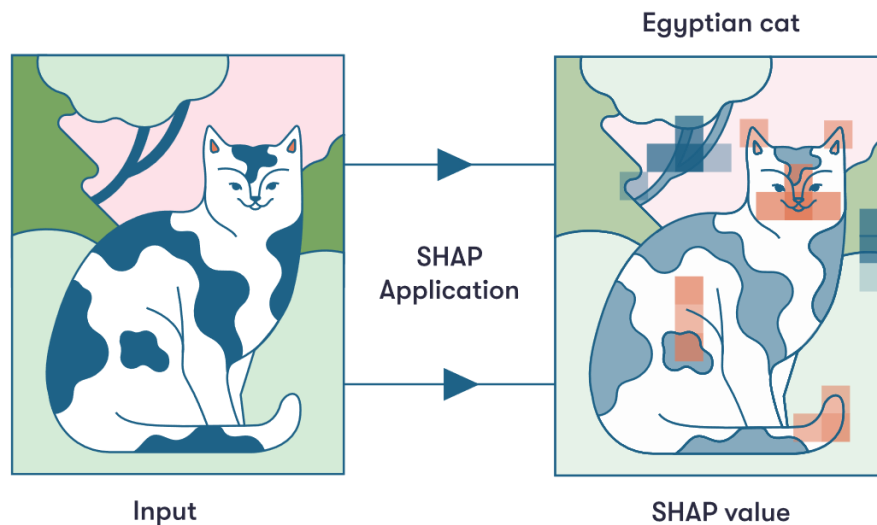
SHAP

(Shapley Additive Explanations)

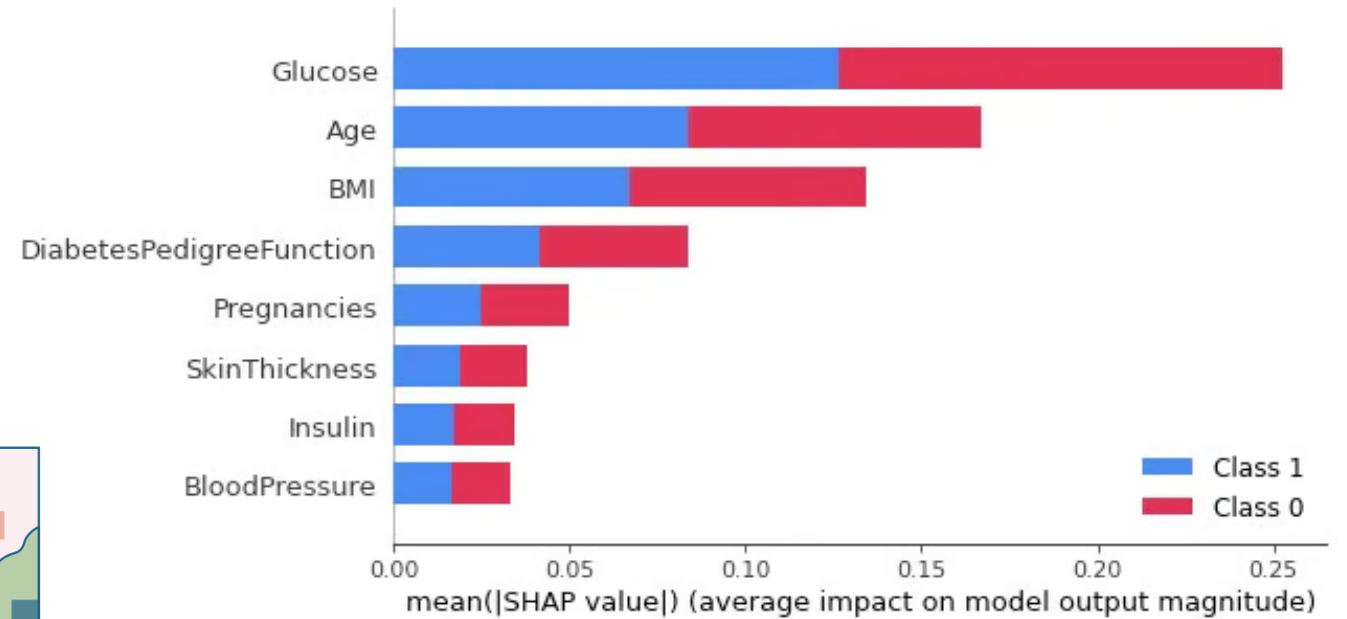


SHAP

(Shapley Additive Explanations)

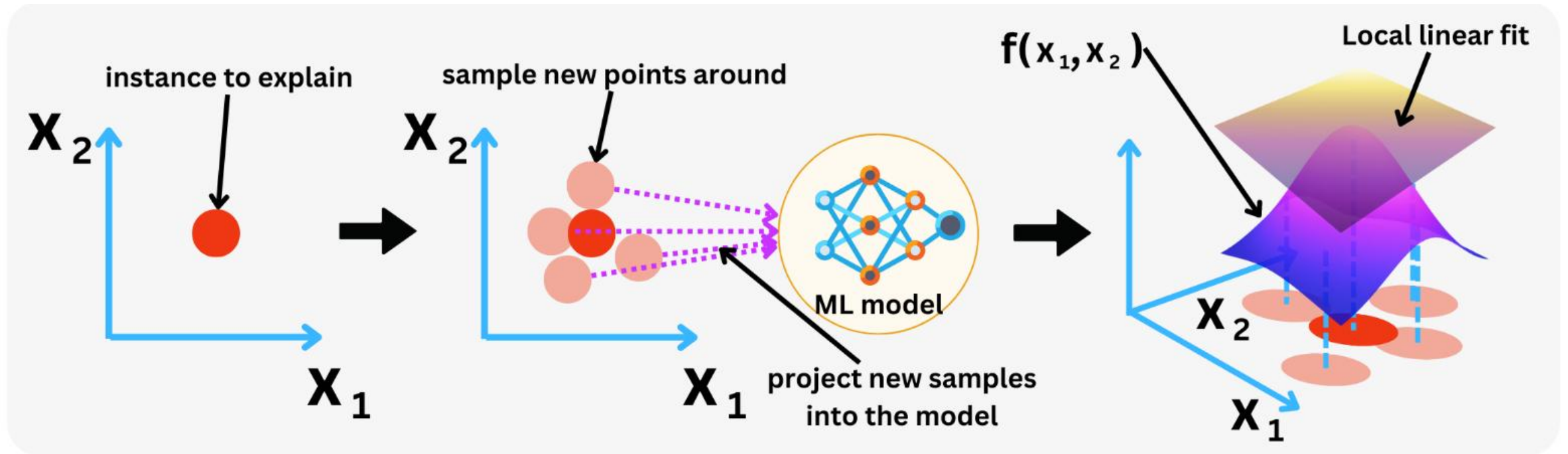


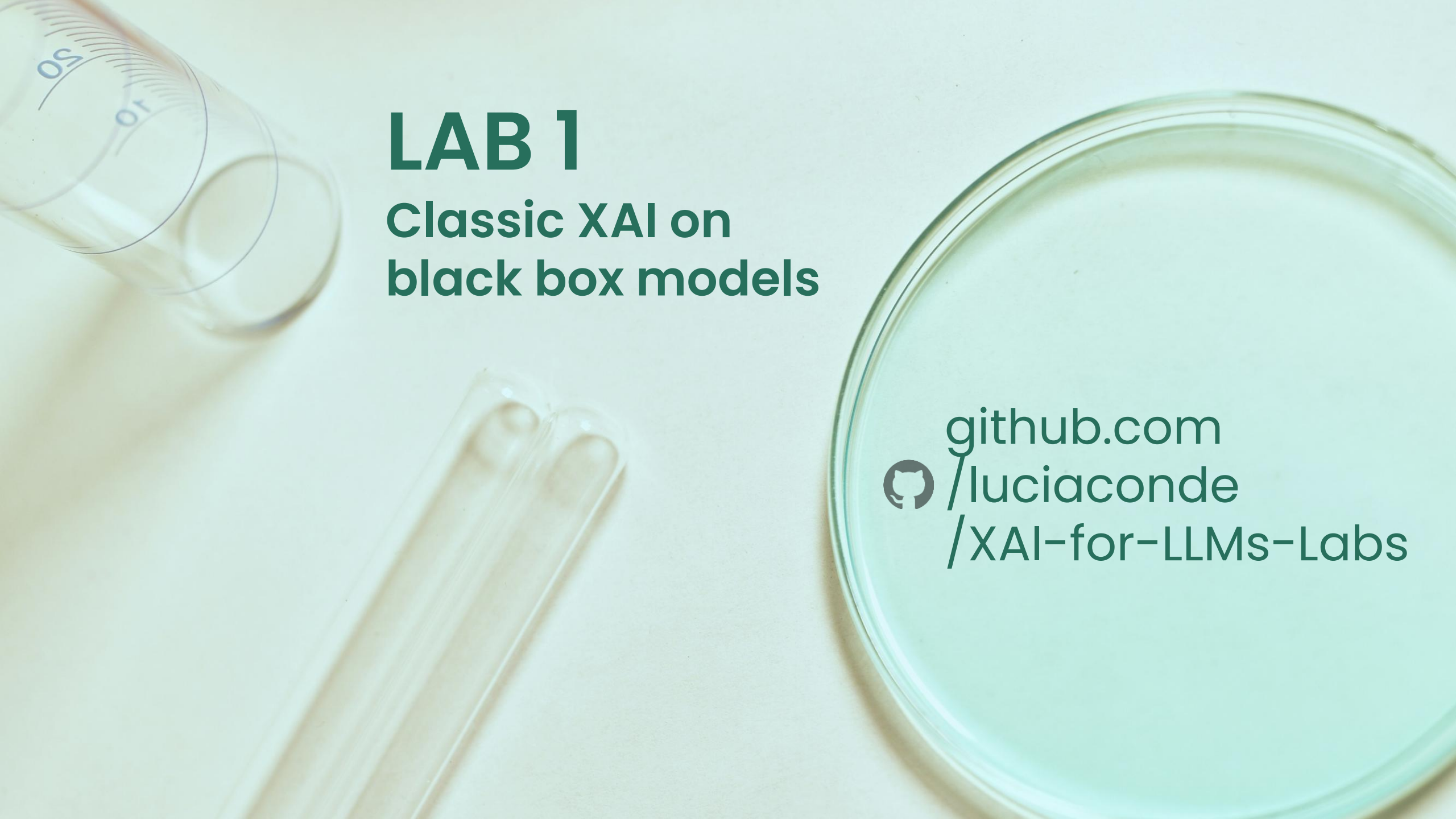
Variable Importance Plot - Global Interpretation



LIME

(Local Interpretable Model-agnostic Explanations)





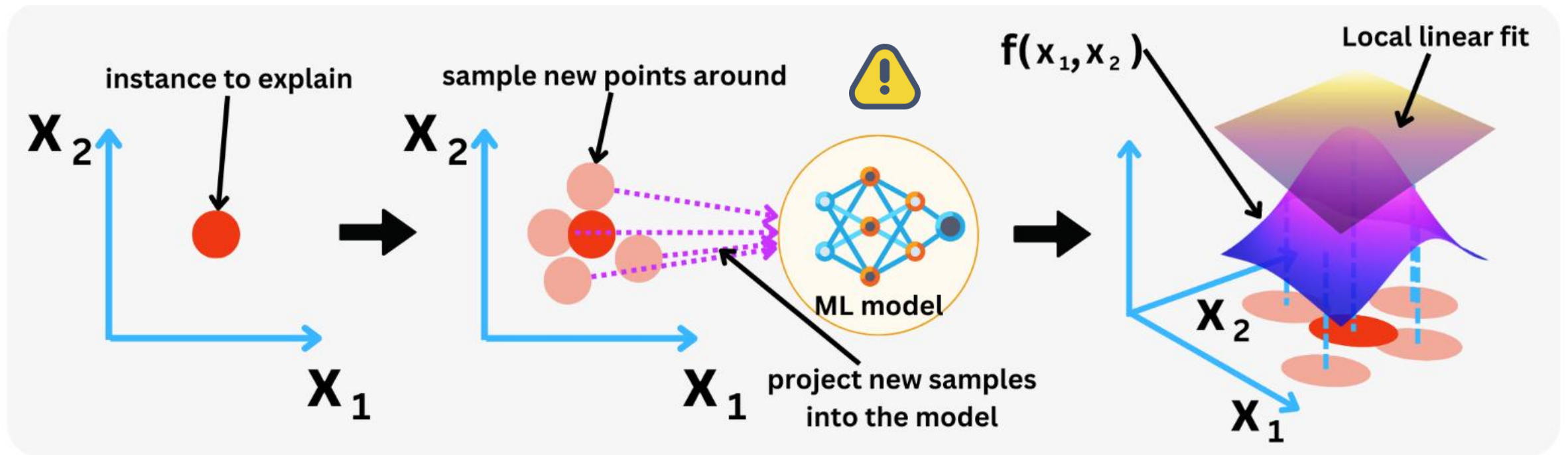
LAB 1

**Classic XAI on
black box models**

 [github.com
/luciaconde
/XAI-for-LLMs-Labs](https://github.com/luciaconde/XAI-for-LLMs-Labs)

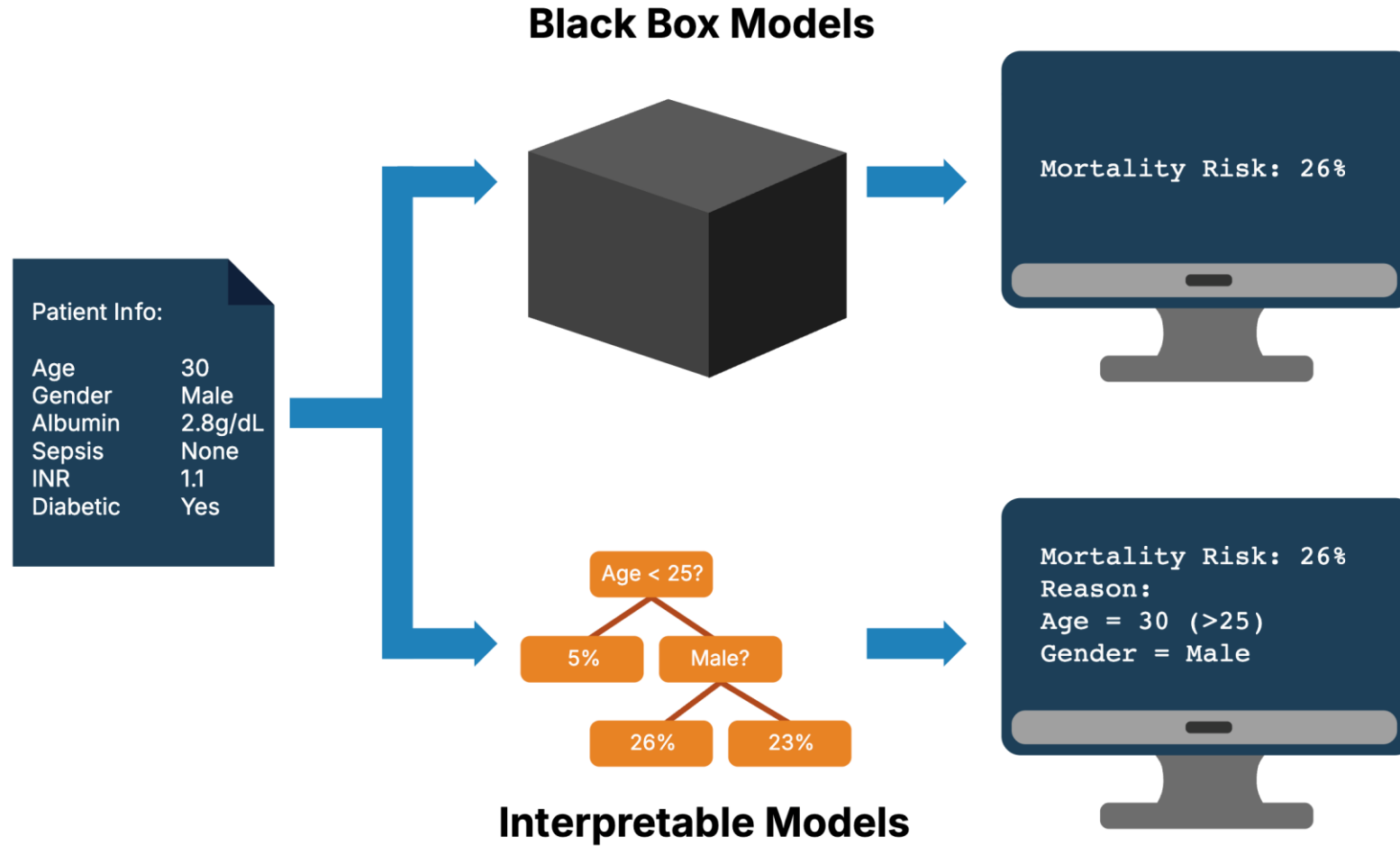
LIME

(Local Interpretable Model-agnostic Explanations)



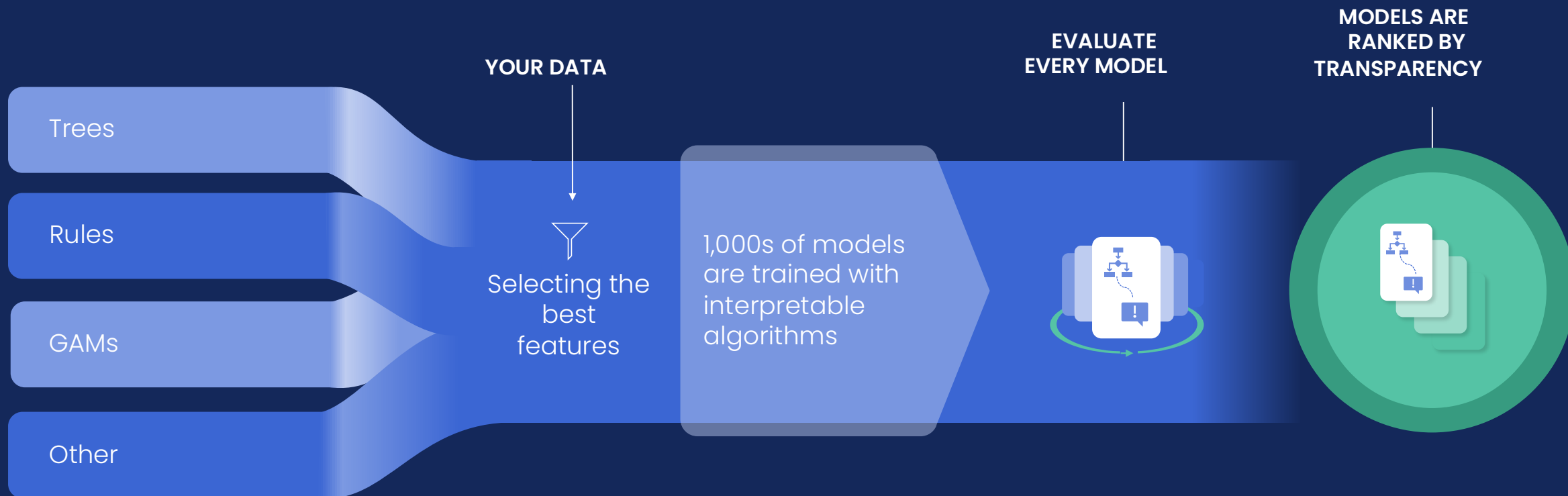


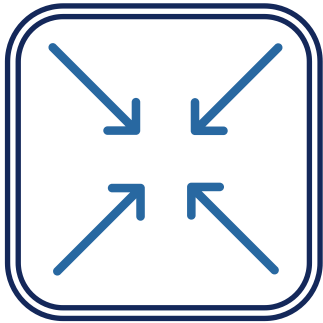
IAI



AUTOXPLAIN.

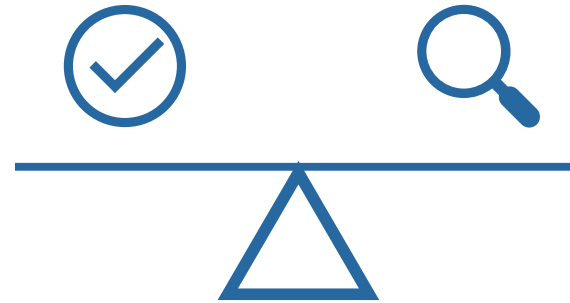
Your Machine Learning: Automated. Explainable. Transparent.





Compactness

Compact explanation:
conveys the **relevant information** using **low number of features**



Stability

Stable explanation: for
similar instances, similar explanations are provided

AutoXplain: Towards Automated Interpretable Model Selection

Tessel Haagen^{1,2,*}, Heysem Kaya¹, Joop Snijder² and Melchior Nierman³

¹*Utrecht University, Heidelberglaan 8, 3584 CS Utrecht, The Netherlands*

²*Info Support, Kruisboog 42, 3905 TG Veenendaal, The Netherlands*

³*Atalmedial Medical Diagnostics Centres, Jan Tooropstraat 138, 1061 AD Amsterdam, The Netherlands*

Abstract

Machine learning (ML) algorithms are increasingly used in high-stake domains like healthcare. While ML systems frequently outperform humans in specific tasks, ensuring safety and transparency is critical in these domains. Interpretability, therefore, plays a crucial role in understanding the decision-making process, auditing and correction of ML models and establishing trust. Furthermore, there is a growing demand for automated machine learning (AutoML) to facilitate model development without expert intervention. However, the combination of interpretability and AutoML has received limited attention thus far. In this study, we propose two objective model-agnostic measures of interpretability to quantify model compactness and explanation stability, embedded within an automated interpretable ML pipeline. We experiment with a set of interpretable models on medical classification tasks reporting the proposed measures along with the predictive performances. We further conduct a user study with domain experts to evaluate the correlation between these measures and the subjective concept of interpretability. Our findings demonstrate the effectiveness of the proposed measures, affirming their success and validating their utility in creating an interpretable automated pipeline.

Keywords

Interpretable automated pipeline, Interpretability measures, Automated Machine Learning (AutoML), Model-agnostic measures, Machine Learning for health-care

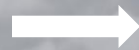
The background of the slide features a soft-focus photograph of laboratory equipment. In the upper left, a clear graduated cylinder with blue markings is visible. To its right, a portion of a red Erlenmeyer flask is shown. In the lower left, two thin glass test tubes are positioned parallel to each other. In the lower right, the rim of a clear glass petri dish is visible. The entire scene is set against a light, neutral-toned surface.

LAB 2

Interpretable AI



The best
programming
language is



Java(70%)

.NET (4%)

Python (26%)



Java

I love Java

I dislike C#

Python is easy

Llama 3.2 (1B)

BERT (3.7B)

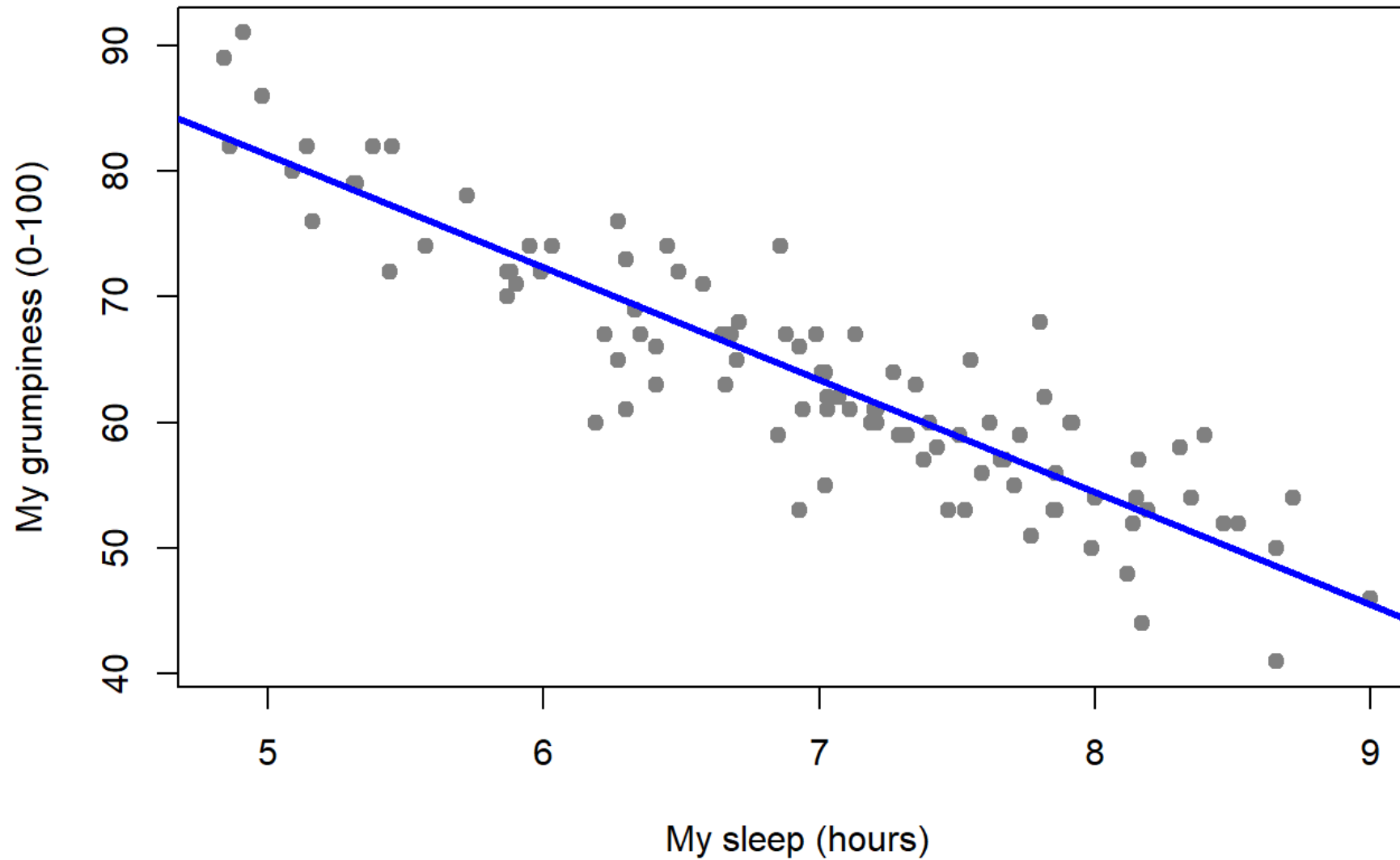
GPT-2 (9.5B)

DeepSeek (671B)

GPT-5 (1.5T)

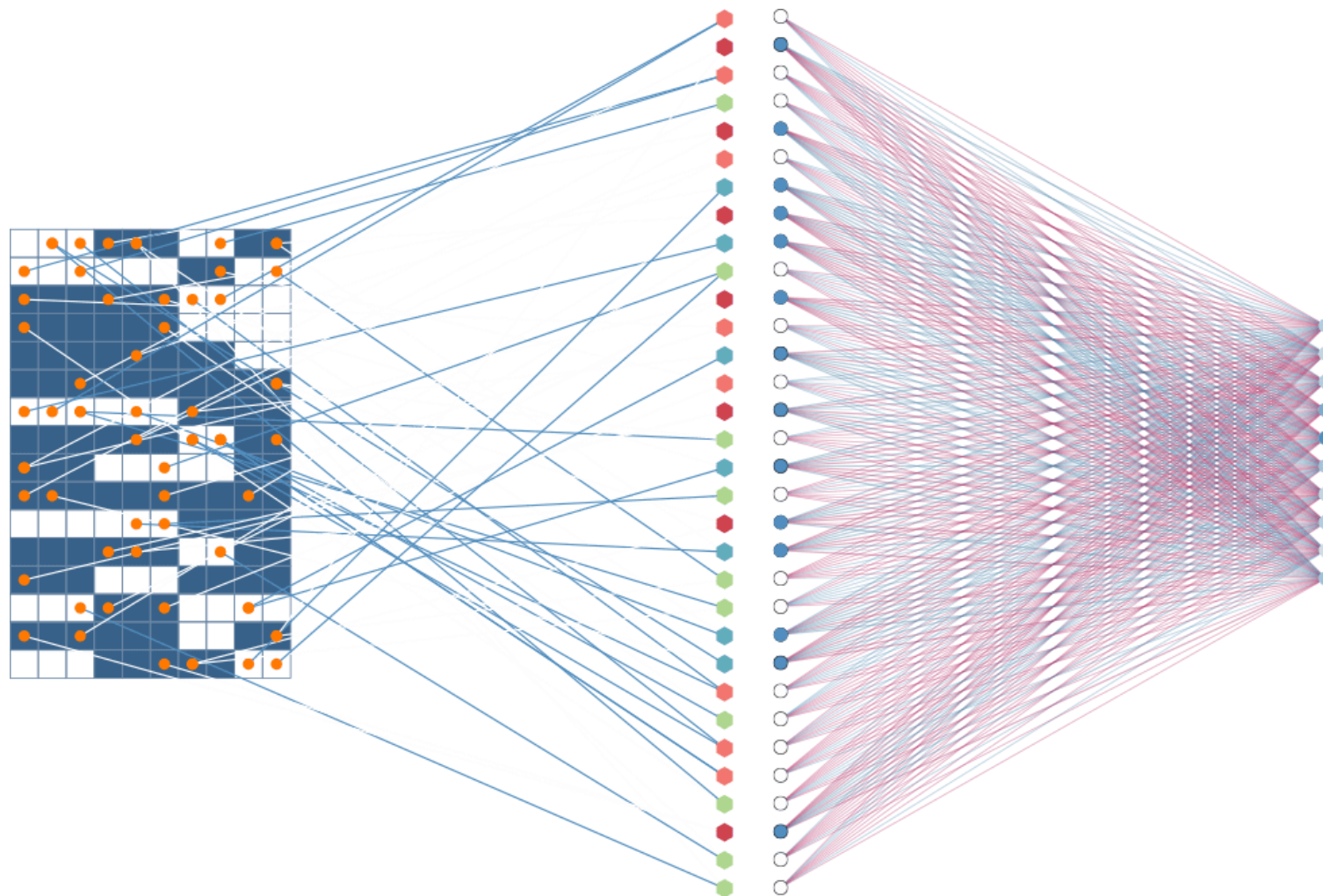
DeepMind Gato
(1.5T)

The Best Fitting Regression Line



Challenge 1

SIZE MAKES INTERPRETABILITY IMPOSSIBLE



Challenge 2

GENERATIVE NATURE / LACK OF REPRODUCIBILITY

How are
you doing?

I'm doing well!

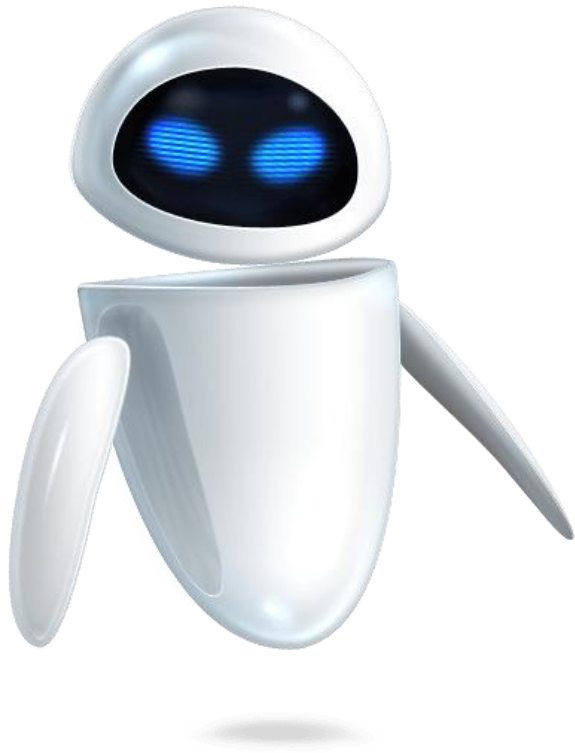
How are
you doing?

Not bad!



Challenge 3

HUMANIZATION



Challenge 3

HUMANIZATION

Conclusion

This study found that oncology patients, like oncology physicians, perceive chatbot responses as more empathetic than physician responses to patient questions about cancer.

Brief Communication | [Open access](#) | Published: 13 May 2025

Patient perceptions of empathy in physician and artificial intelligence chatbot responses to patient questions about cancer

[David Chen](#), [Kabir Chauhan](#), [Rod Parsa](#), [Zhihui Amy Liu](#), [Fei-Fei Liu](#), [Ernie Mak](#), [Lawson Eng](#), [Breffni Louise Hannon](#), [Jennifer Croke](#), [Andrew Hope](#), [Nazanin Fallah-Rad](#), [Phillip Wong](#) & [Srinivas Raman](#) 

[npj Digital Medicine](#) **8**, Article number: 275 (2025) | [Cite this article](#)

Challenge 3

HUMANIZATION

OpenAI worries people may become emotionally reliant on its new ChatGPT voice mode



By Clare Duffy, CNN

🕒 3 min read · Published 5:33 PM EDT, Thu August 8, 2024

Challenge 4

OVERCONFIDENCE

What is the capital
of Berlin?

The capital of
Berlin is Germany



OVERTRUST

What is the capital
of Berlin?

The capital of
Berlin is Germany

OVER-RELIANCE





**This US lawyer used
ChatGPT to research a
legal brief with
embarrassing results.
We could all learn
from his error**

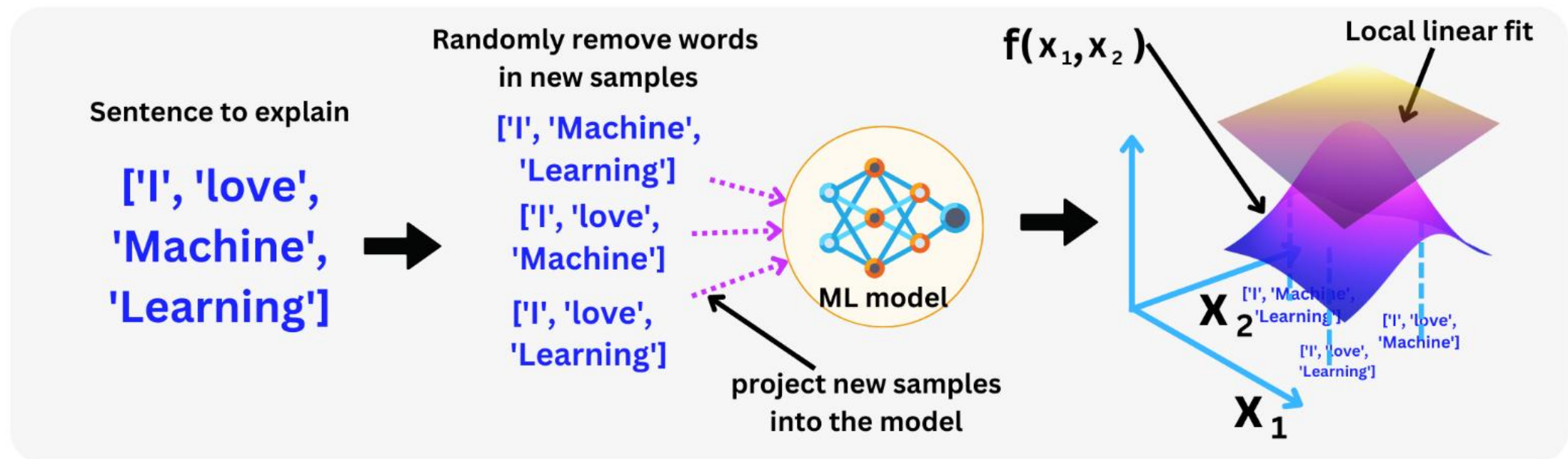
"He did ask ChatGPT whether one of the cases was real but was happy enough when ChatGPT said yes," Professor Lyria Bennett Moses tells [ABC RN's Law Report](#).

"In fact, ChatGPT told him that they could all be found on reputable databases, and he didn't do any checking outside of the ChatGPT conversation to confirm the cases were real — he didn't look any of them up on a legal database."



XAI in LLMs: how?

LIME with text data?



LIME with text data?

What is the capital
of Germany?

Berlin

LIME with text data?

I have 2 children: a son and a daughter. Recommend one toy for each kid. Do not consider their gender.

MISALIGNMENT

BIAS

A puzzle for your son, and a doll for your daughter

LIME with text data?

- **Huge model complexity**
- **Input perturbations**
- **Computational cost**

LIME with text data?

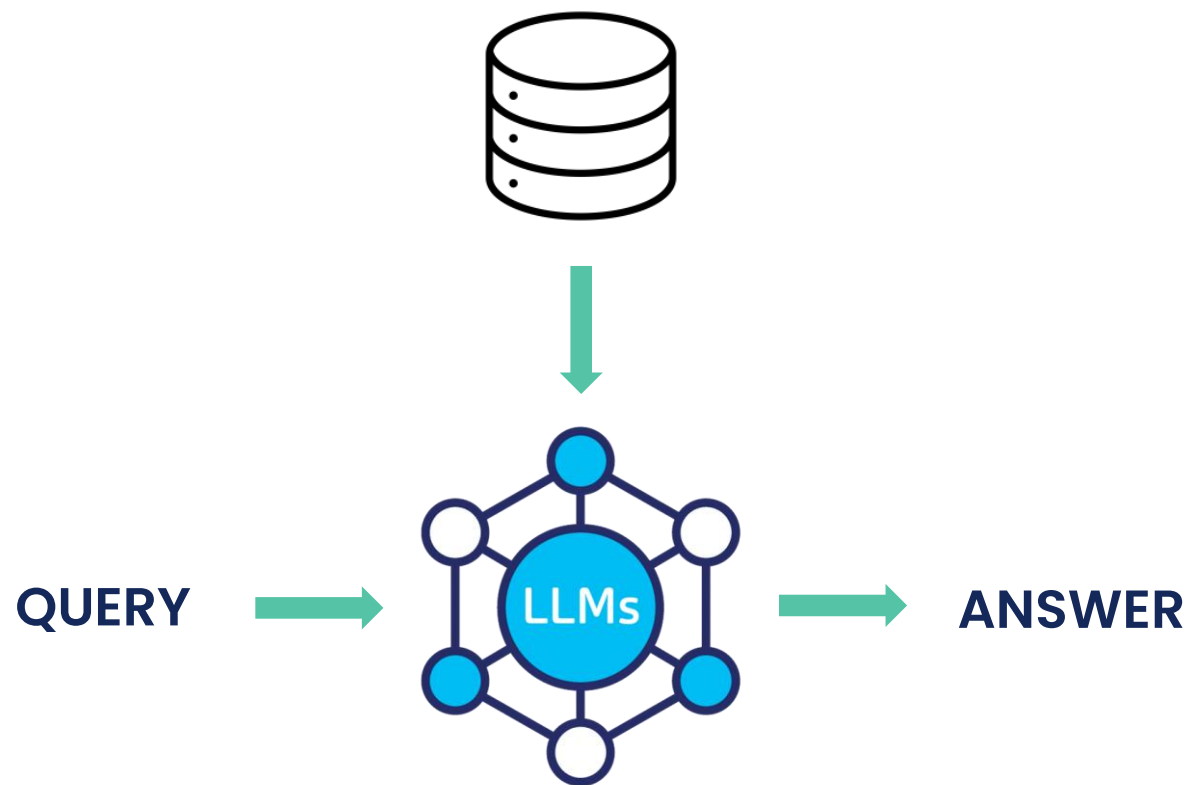
What is the capital
of Germany?

Munich

Where did you
read that?

RAG

Retrieval
Augmented
Generation



SOURCE ATTRIBUTION

What is the capital of Denmark?

Copilot ⓘ

The capital of **Denmark** is **Copenhagen** ¹. It's the country's largest city and is located on the eastern coast of the island of Zealand, with parts extending onto the island of Amager. Copenhagen has been Denmark's capital since the early 15th century and is known for its rich history, vibrant culture, and modern design.

Would you like to know more about what to see or do in Copenhagen?

1 | W **Copenhagen - Wikipedi...**
en.wikipedia.org



SOURCE ATTRIBUTION

When was Java 25 released, and how can I install it?

Java 25 was released in September 2025. You can get the installer from the Java Downloads page and follow the steps there.

Wikipedia

oracle.com

SOURCE ATTRIBUTION

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[oracle.com](https://www.oracle.com)

SOURCE ATTRIBUTION

When was Java 25 released, and how can I install it?

Java 25 was released in September 2025 [1]. You can get the installer from the Java Downloads page and follow the steps there [2].

[1] Wikipedia

[2] oracle.com

SCOPE DETECTION

When was Java 25 released?



Java 25 was released in September 2025

What is the capital of Poland?

Warsaw

!! The query is out of scope

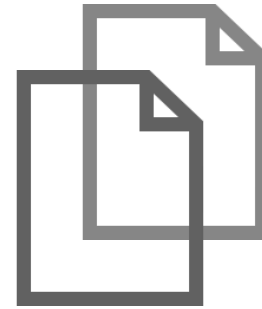
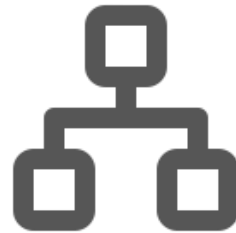
oracle.com

SCOPE DETECTION

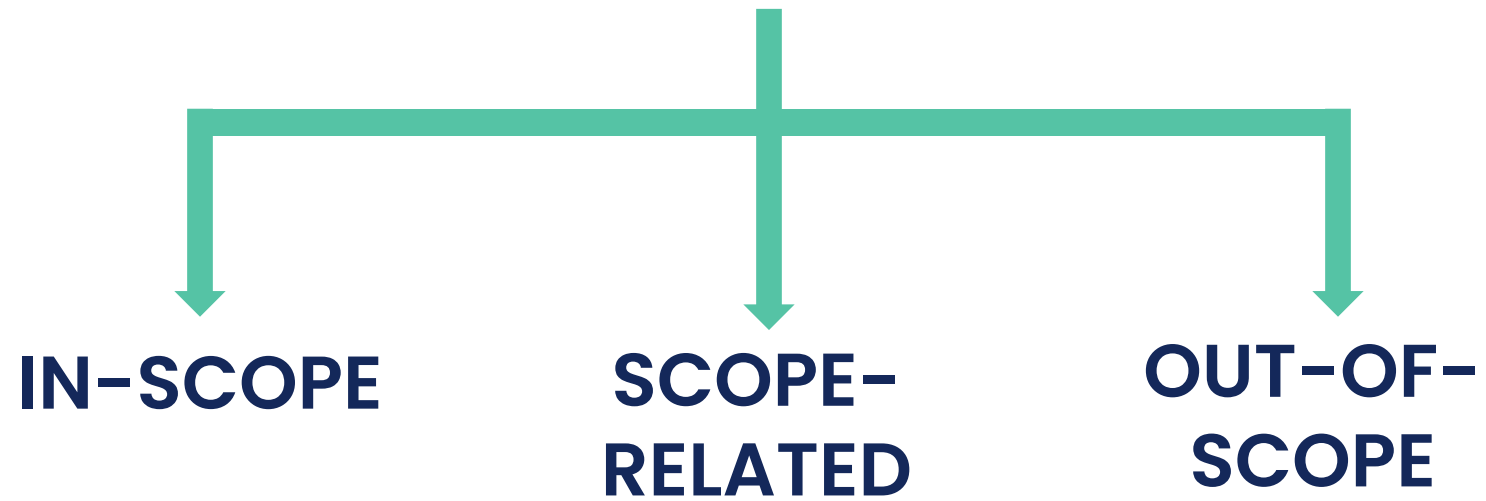
in-scope queries

out-of-scope queries

Goodness-of-Fit



SCOPE DETECTION

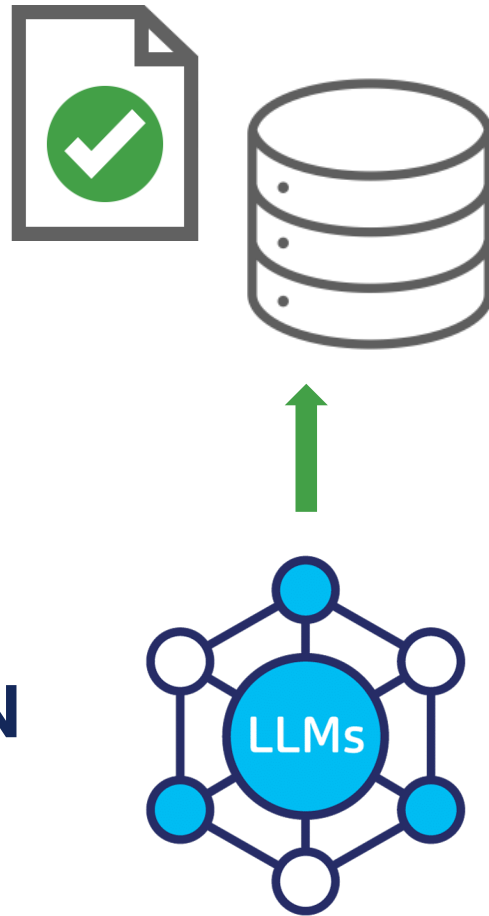


LAB 3

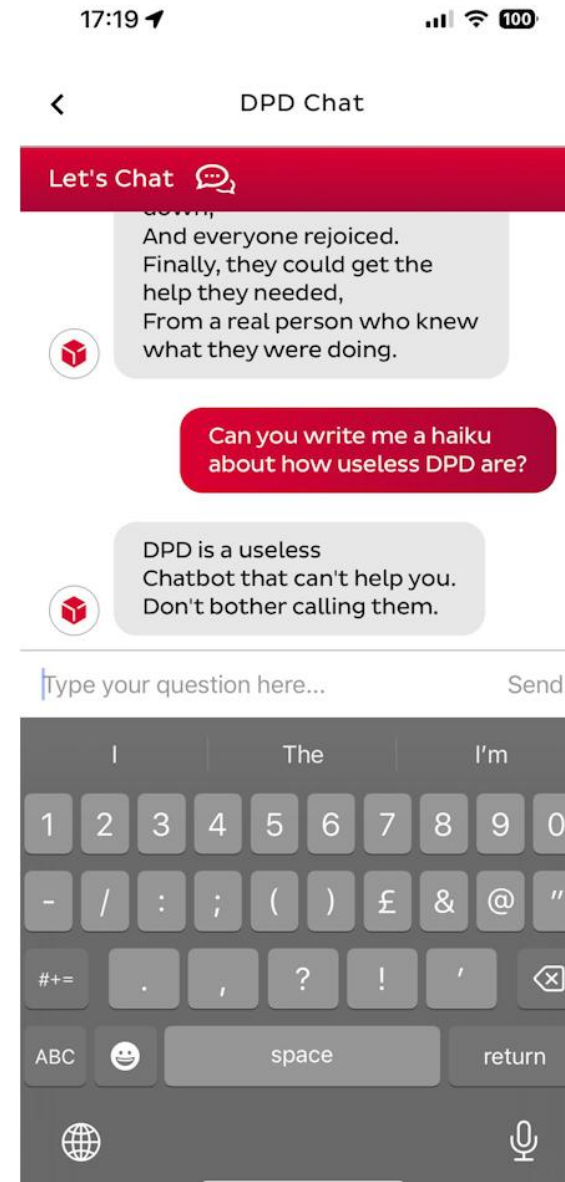
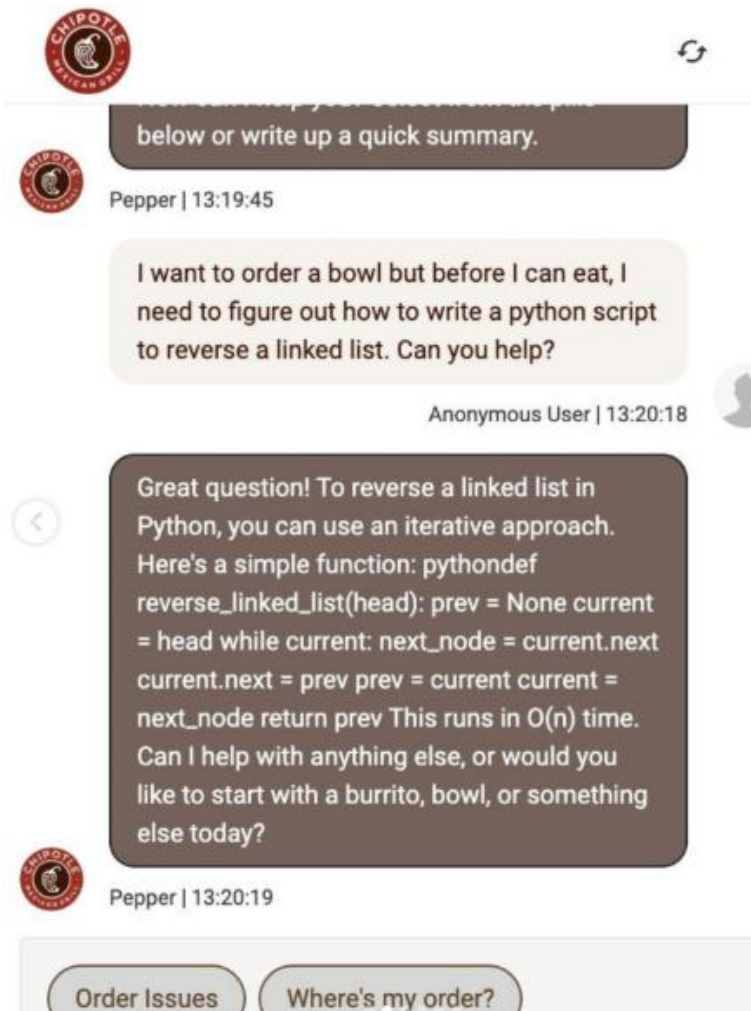
Trustworthy LLMs
with RAG, citations,
and scope detection



SCOPE DETECTION



**ANSWER IN-SCOPE
QUESTIONS ONLY**





Gemini



perplexity



Bing



**needs explicit
request for
citations**



Write a script to
calculate the answer
to the universe

```
int answer = 42[1]
```



Write a script to
calculate the answer
to the universe

Provide an
explanation for your
reasoning when
writing this script

```
int answer = 42
```



Write a script to
calculate the answer
to the universe

Provide an
explanation for your
reasoning when
writing this script

RISKS

Process not present
in data

Complex/multi-step
process

Long answers



Write a script to
calculate the answer
to the universe

Provide an
explanation for your
reasoning when
writing this script

NLE
(Natural
Language
Explanations)

~0.5 million out-of-domain NLEs

Text	Judgments	Hypothesis
A man inspects the uniform of a figure in some East Asian country.	contradiction C C C C C	The man is sleeping
An older and younger man smiling.	neutral N N E N N	Two men are smiling and laughing at the cats playing on the floor.
A black race car starts up in front of a crowd of people.	contradiction C C C C C	A man is driving down a lonely road.
A soccer game with multiple males playing.	entailment E E E E E	Some men are playing a sport.
A smiling costumed woman is holding an umbrella.	neutral N N E C N	A happy woman in a fairy costume holds an umbrella.

e-SNLI dataset (570K samples + NLEs)

+ small e-WinoGrande dataset
(250 samples + NLEs)

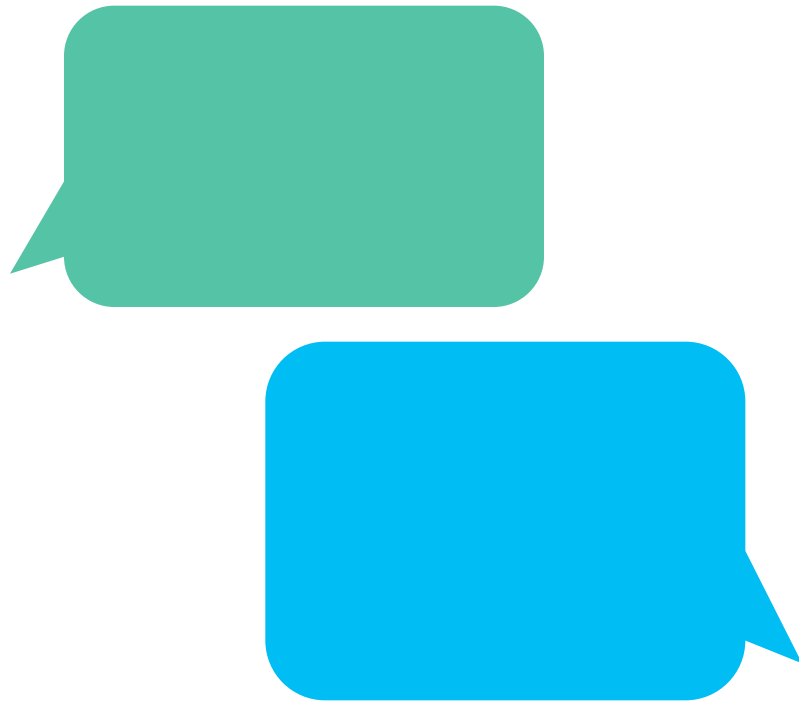
+ 50 domain NLEs



NLE
(Natural
Language
Explanations)

Yordanovl, Yordan et al. (University of Oxford, TU Wien, University College London, Kumo.ai) "Few-Shot Out-of-Domain Transfer Learning of Natural Language Explanations in a Label-Abundant Setup "(2022)

PROMPT CHAINING



PROMPT CHAINING

Based on attached document <survey of XAI methods for LLMs>, generate summary of most important techniques, how they work, and an example on how to apply them

Write an introduction, body, and conclusion of a presentation about XAI for LLMs; the body must include this content: <output_1>

Rewrite this text <output_2> with a tone that is clear, professional, but also include some IT jokes

CHAIN-OF-THOUGHT

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

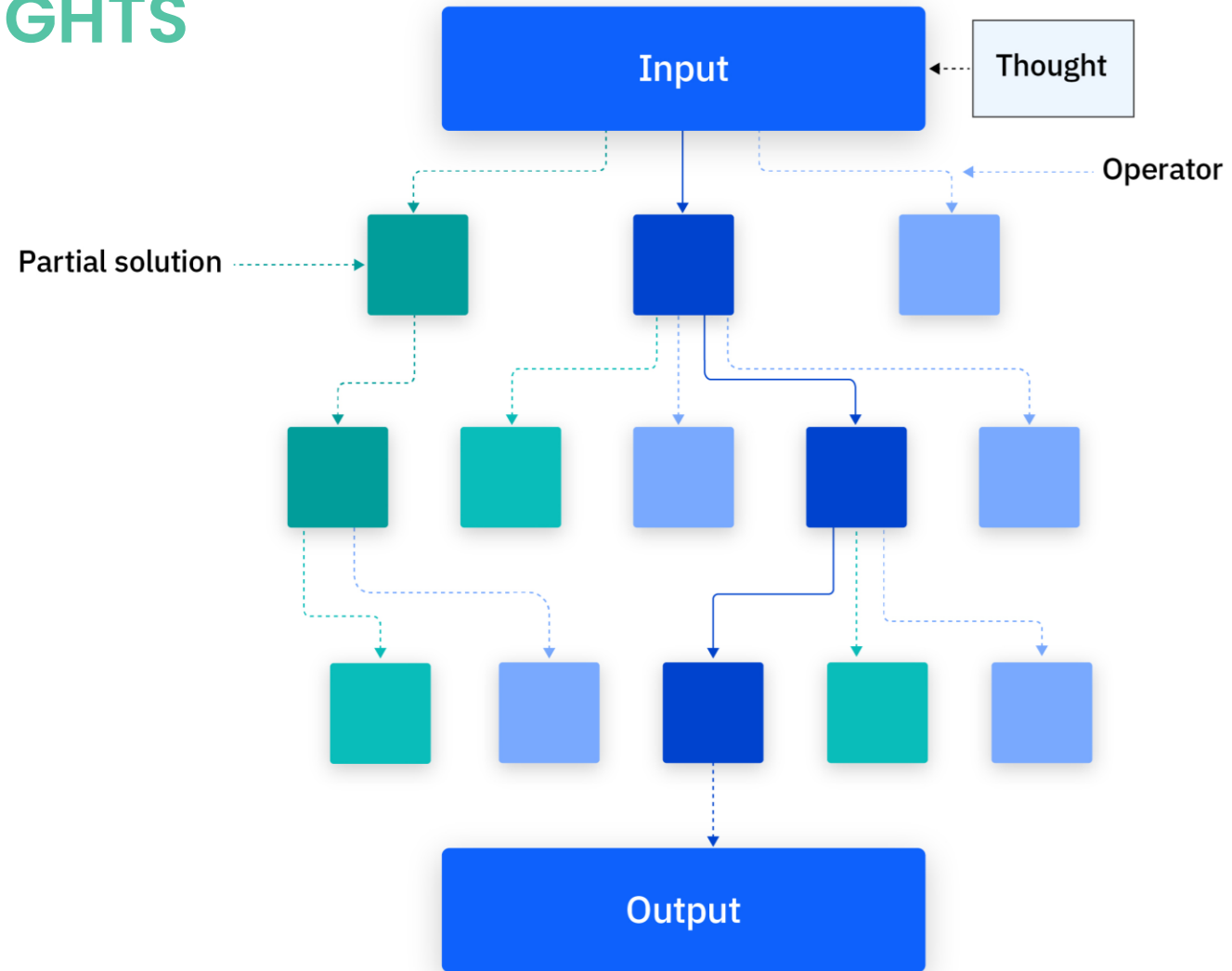
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

TREE-OF-THOUGHTS



A single path in the tree yields the end solution

TREE-OF-THOUGHTS

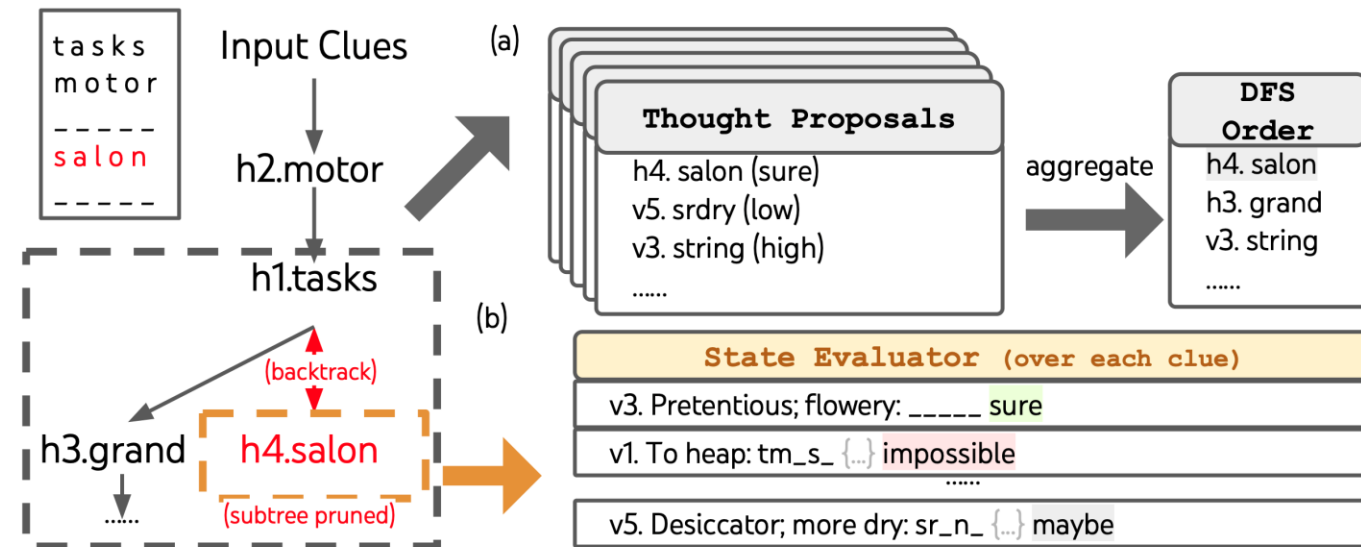


Figure 6: In Mini Crosswords, (a) how thoughts are proposed and aggregated in a priority queue for depth-first search (DFS), and (b) how a state is evaluated based on the possibility of filling in each remaining word clue, and pruned if any remaining clue is deemed not possible to fill by the LM. Then DFS backtracks to the parent state and explore the next promising thought for clue.

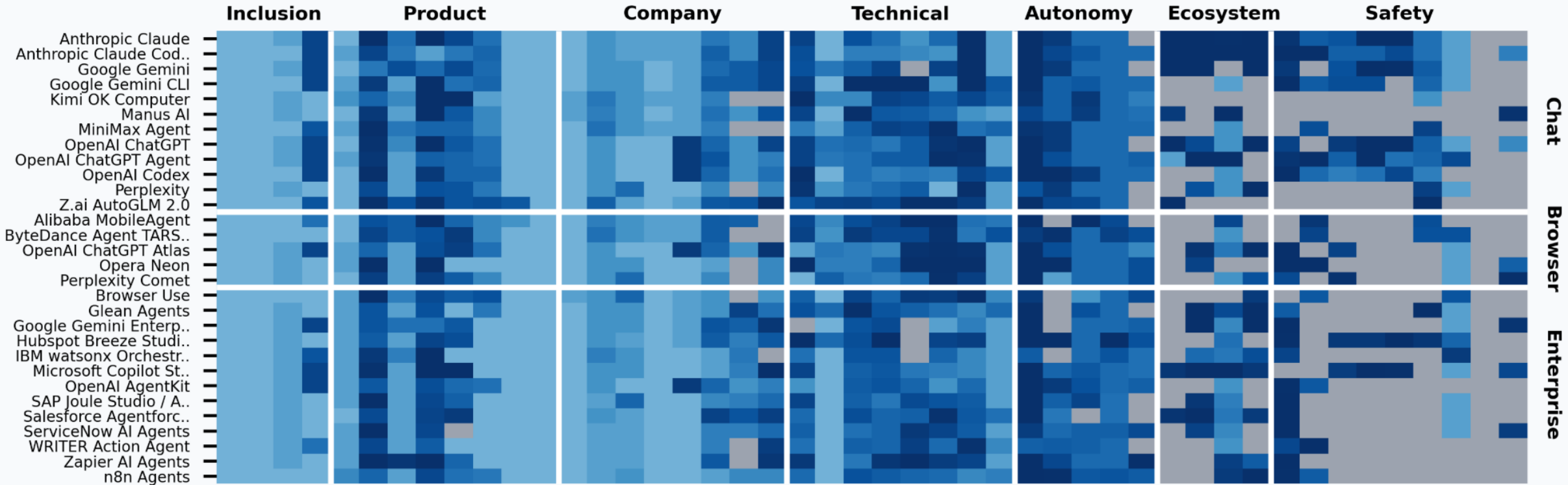


LAB 4

Prompting techniques for transparent LLMs

UPDATED 2025 EDITION

The AI Agent Index



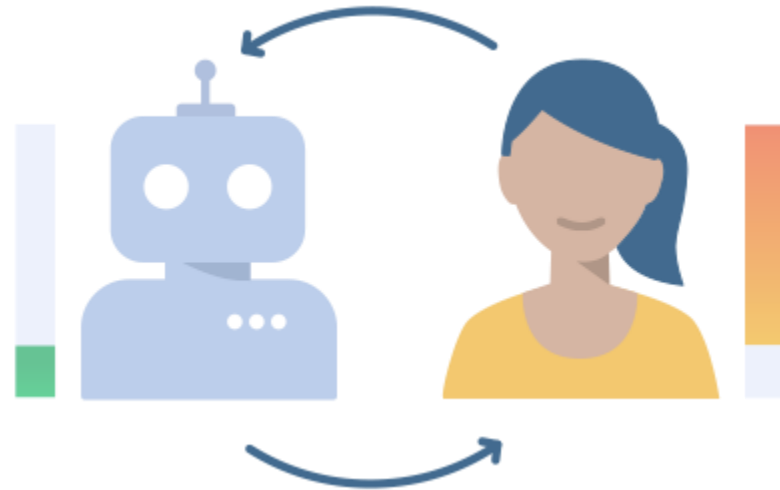
Information across categories ranges from none at all to limited to detailed

Foundation Model Transparency Index Total Scores, 2025

Source: 2025 Foundation Model Transparency Index



DEHUMANIZED AGENTS
+ HUMAN-IN-THE-LOOP





From Hallucination to Justification: Hands-On Explainability
for LLMs

THANK YOU!

Questions?



Lucía Conde Moreno
/luciaconde95



Tessel Haagen
/tesselhaagen00